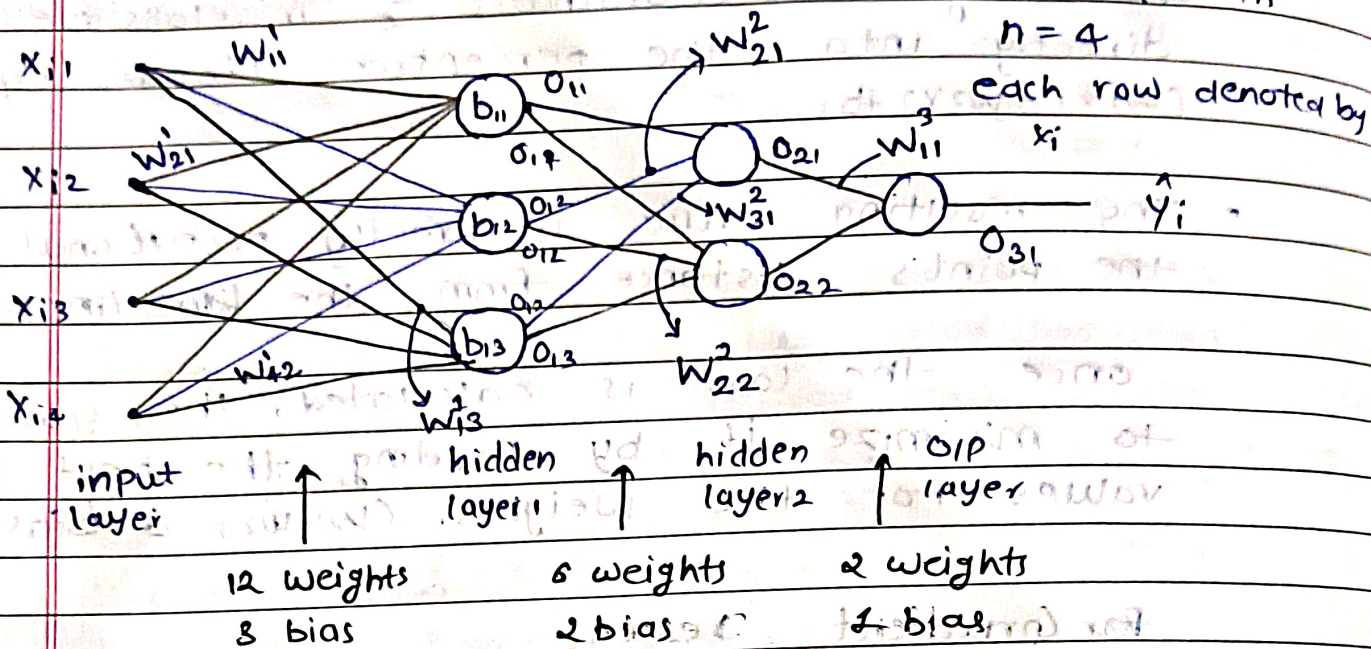


Multi Layer Perceptron

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MLP [multilayer perceptron] Notation.



Total trainable parameters: 26

In MLP, how weights, bias & outputs are denoted?

bias - b_{ij}

i - layer number

j - Node number in that layer

ex, b_{12} - 1st layer, 2nd node.

output - O_{ij}

weights - W_{ij}^k

k - weight कोणसे layer में घूस रहा है

i - current layer में कोणसे नंबर वाले नोड में निकल रहा है

j - जिस layer में घूस रहा है उस layer में कोणसे नंबर वाले नोड पे घूस रहा है

Loss Function.

Loss function is a method of evaluating how well your algorithm is modeling your dataset.

If output of loss function is

- high $\rightarrow$ algorithm is performing poorly
- small $\rightarrow$ — great.

- It is basically a mathematical function.
- ये एक मशीन learning के algorithms के parameters का function होता है.
- So, if adjust the parameters, the values loss function will change.

Why is Loss Function important?

[you can't improve what you can't measure]

Loss function in DL

ex., we are using MSE

- ① based on the values of weights & bias, we will calculate the output.
- ② then find loss, i.e. $(y_i - \hat{y})^2$
- ③ Using gradient descent, we will update the values of weights & bias.

• We will continue this process for multiple epochs, and in each epoch, the loss will get reduced.

• The point, at which we will get the minimum loss, our model will predict the output more accurately.

Loss Function.

In ML, Loss function is a way to tell how good or bad the machine learning model is performing.

loss funcⁿ for perceptron:

$$L = \frac{1}{n} \sum_{i=1}^n \max(0, -y_i f(x_i))$$

rows in data

y_i - constant

$$f(x_i) = w_1 x_1 + w_2 x_2 + b$$

value of

we have to find $\hat{w}_1, \hat{w}_2, \hat{b}$ such that the loss function will be minimum.

for that we use gradient descent.

Hinge Loss

Explanation of Loss function.

$$L = \frac{1}{n} \sum_{i=1}^n \max(0, -y_i f(x_i))$$

where

$$f(x_i) = w_1 x_{i1} + w_2 x_{i2} + b$$

x_1	x_2	y
x_{11}	x_{12}	y_1
x_{21}	x_{22}	y_2

x_{ij}

row call

$$\max(0, -y_i f(x_i))$$

$$= \max(0, x)$$

consider this as x

$$= x \text{ if } x \geq 0$$

$$= 0 \text{ if } x < 0$$

consider we have 2 points

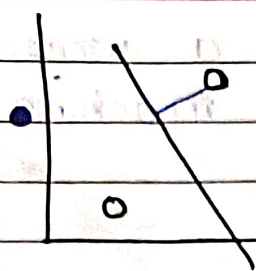
$$L = \frac{1}{2} [\max(0, -y_1 f(x_1)) + \max(0, -y_2 f(x_2))]$$

Geometric meaning of

loss function.

ex.,

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cgpa	Placed	
8	1	● - placed
6	-1	○ - Not placed
4	2	●
1	1	○

y_i $\hat{y}_i$

1 1 $y_i = 1$ $f(x_i) \rightarrow +ve$ (as correctly predicted)
 -1 -1 $\rightarrow$ actual - Not placed, predicted - Not placed
 1 -1 $\rightarrow$ actual - placed, predicted - Not placed
 -1 1 $\rightarrow$ actual - Not placed, predicted - placed

(I) $y_i = 1$ $\hat{y}_i = 1$

$y_i = 1$ $f(x_i) = +ve$

so, $\max(0, -y_i f(x)) \Rightarrow -ve$

$\max(0, -y_i f(x))$ selects 0

this tells there is 0 loss.

(II) $y_i = -1$ $\hat{y}_i = -1$

$y_i = -1$ $f(x_i) = -ve$ (as correctly predicted -ve point)

$-y_i f(x) = -[(-1)(-ve \text{ value})] \Rightarrow -ve$

$\therefore \text{loss} = 0$

(III) $y_i = 1$ $\hat{y}_i = -1$

$y_i = 1$ $f(x_i) = -ve$ (predicted as Not placed)

+ve -ve = -ve

$-y_i f(x) = -[1(-ve \text{ value})] \Rightarrow +ve$

$\max(0, -y_i f(x))$ select the result of $[-y_i f(x)]$

(IV) $y_i = -1$ $\hat{y}_i = 1$

y_i $f(x_i)$ product $-(y_i f(x_i))$

-ve +ve -ve $\Rightarrow +ve$

Basically,

In this loss function,

- we plug the coordinates of misclassified points directly into the perceptron's linear equation $(w_1x_1 + w_2x_2 + b)$
- The resulting value is directly proportional to the point's distance from the line.

once the loss is calculated, the goal is to minimize it by finding the best possible values for the weights (w_1, w_2) & bias (b) .

for Gradient Descent

for i in epochs

$$w_1 = w_1 + \eta \frac{\partial L}{\partial w_1}$$

$$w_2 = w_2 + \eta \frac{\partial L}{\partial w_2}$$

$$b = b + \eta \frac{\partial L}{\partial b}$$

Loss function	Activation	output
Hinge Loss	step	perceptron → binary classification
log-loss (binary cross entropy)	sigmoid	logistic regression → binary classification
categorical cross entropy	softmax	softmax regression → multiclass classification
MSE	Linear	Linear regression o/p: number